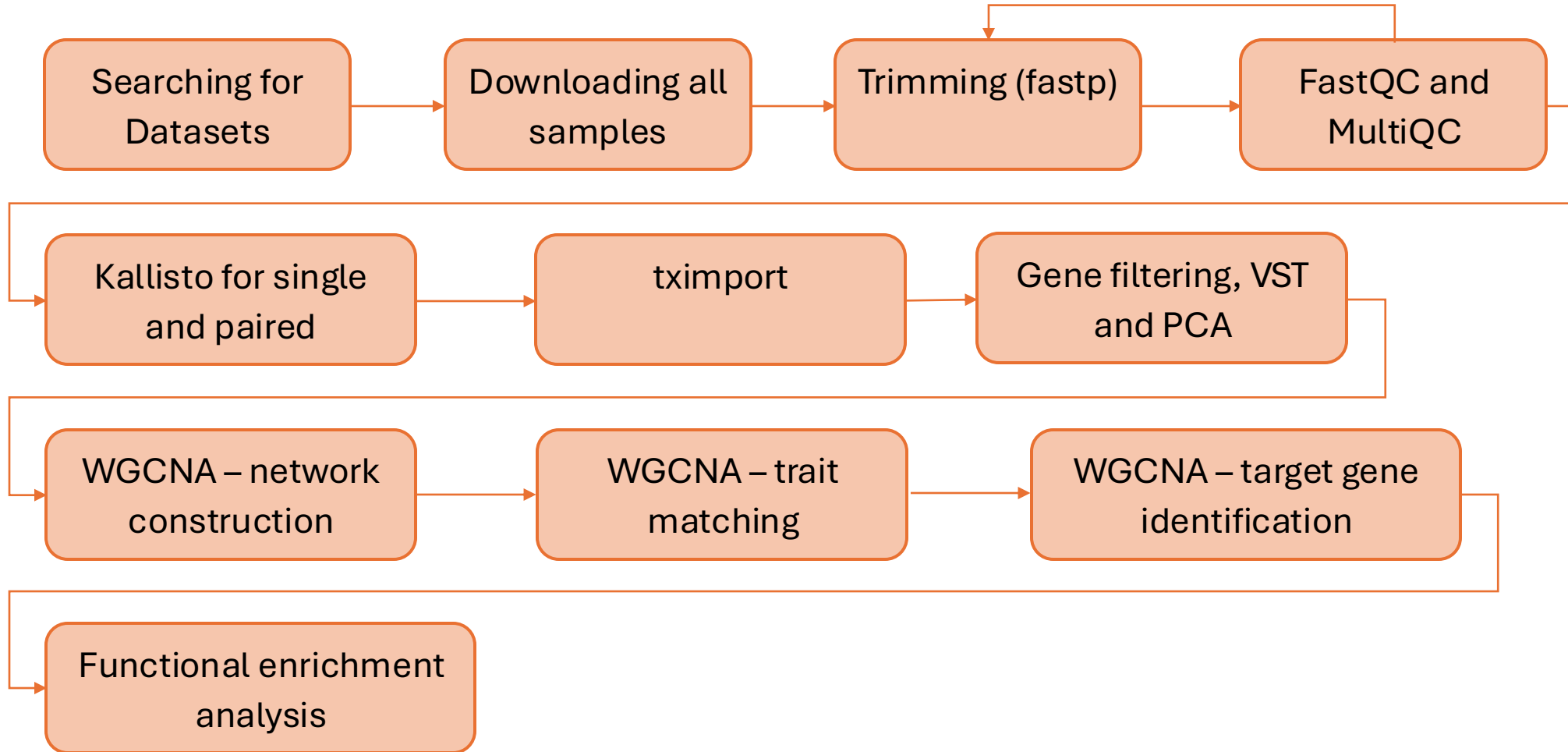


Progress



Datasets

Biotrophic pathogens:

- *Golovinomyces orontii*
- *Erysiphe cichoracearum*
- *Hyaloperonospora arabidopsidis* (2 Datasets)
- *Plasmodiophora brassicae* (2 Datasets)
- Turnip mosaic virus (TuMV) (2 Datasets)
- *Heterodera schachtii*
- Turnip yellows virus (TuYV)
- Cauliflower mosaic virus (CaMV)
- *Meloidogyne javanica*

Control:

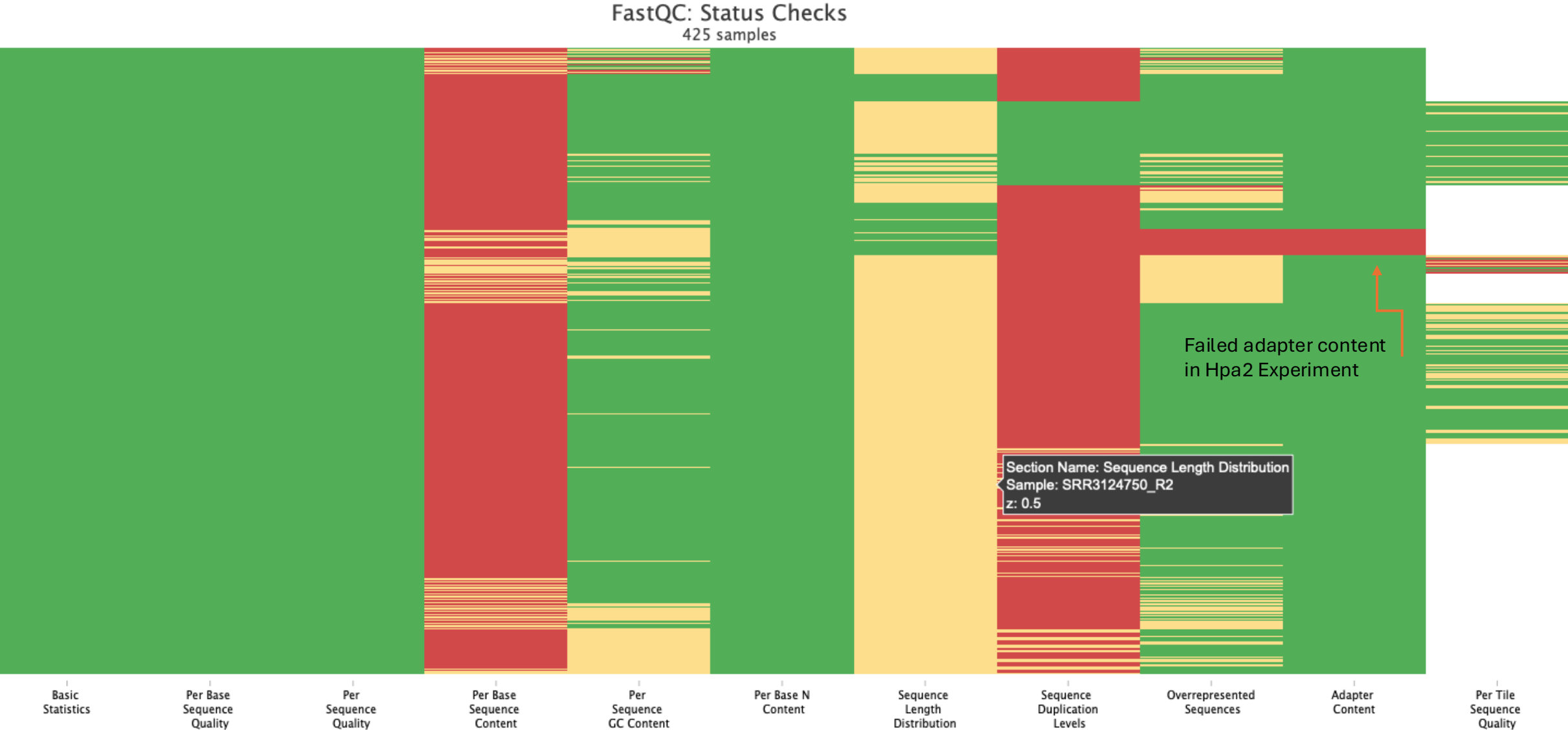
- Drought
- *Botrytis cinerea* (necrotrophic)
- *Alternaria brassicae* (necrotrophic)

Altogether:

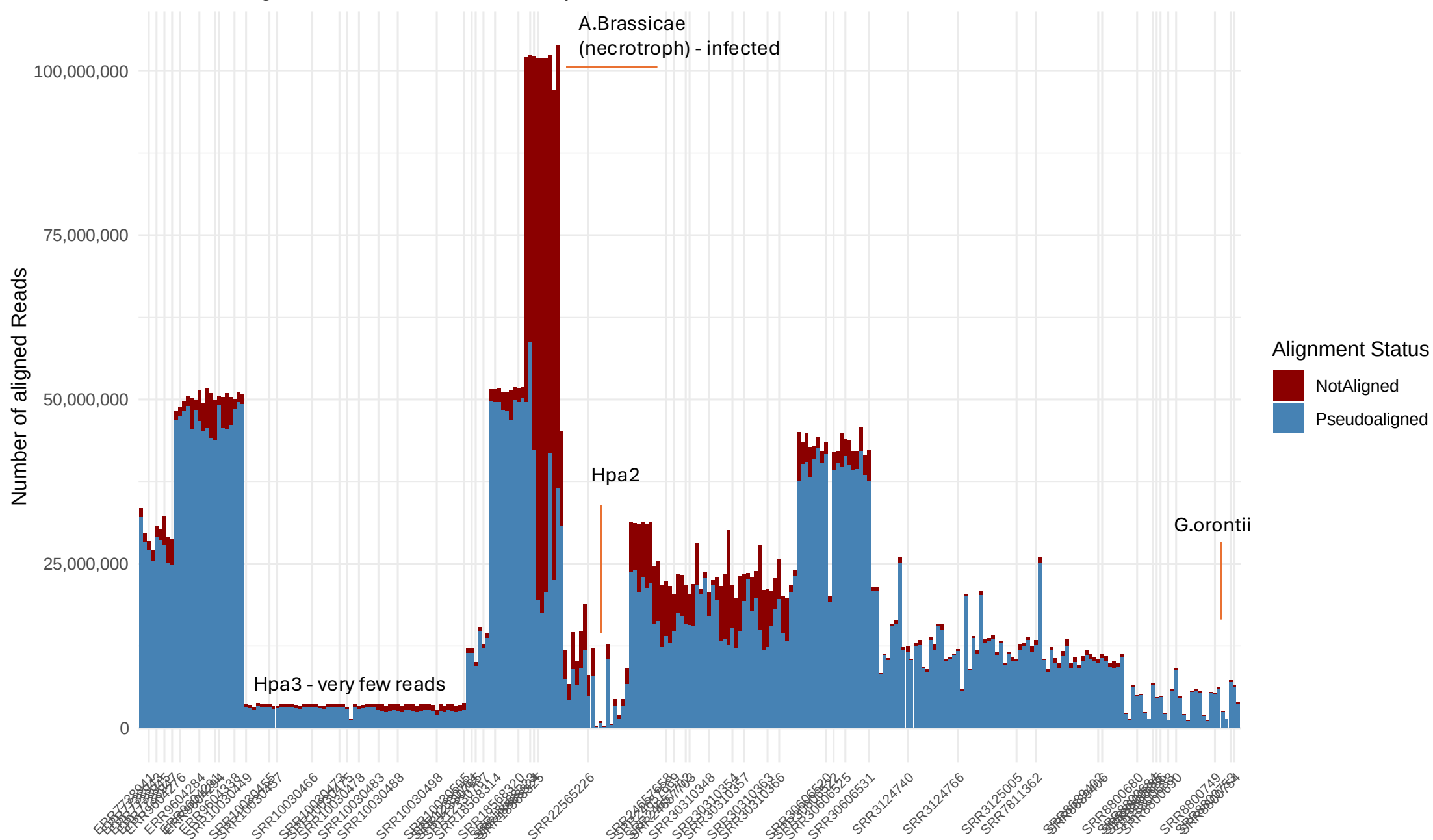
283 samples

141 single-end, 142 paired-end

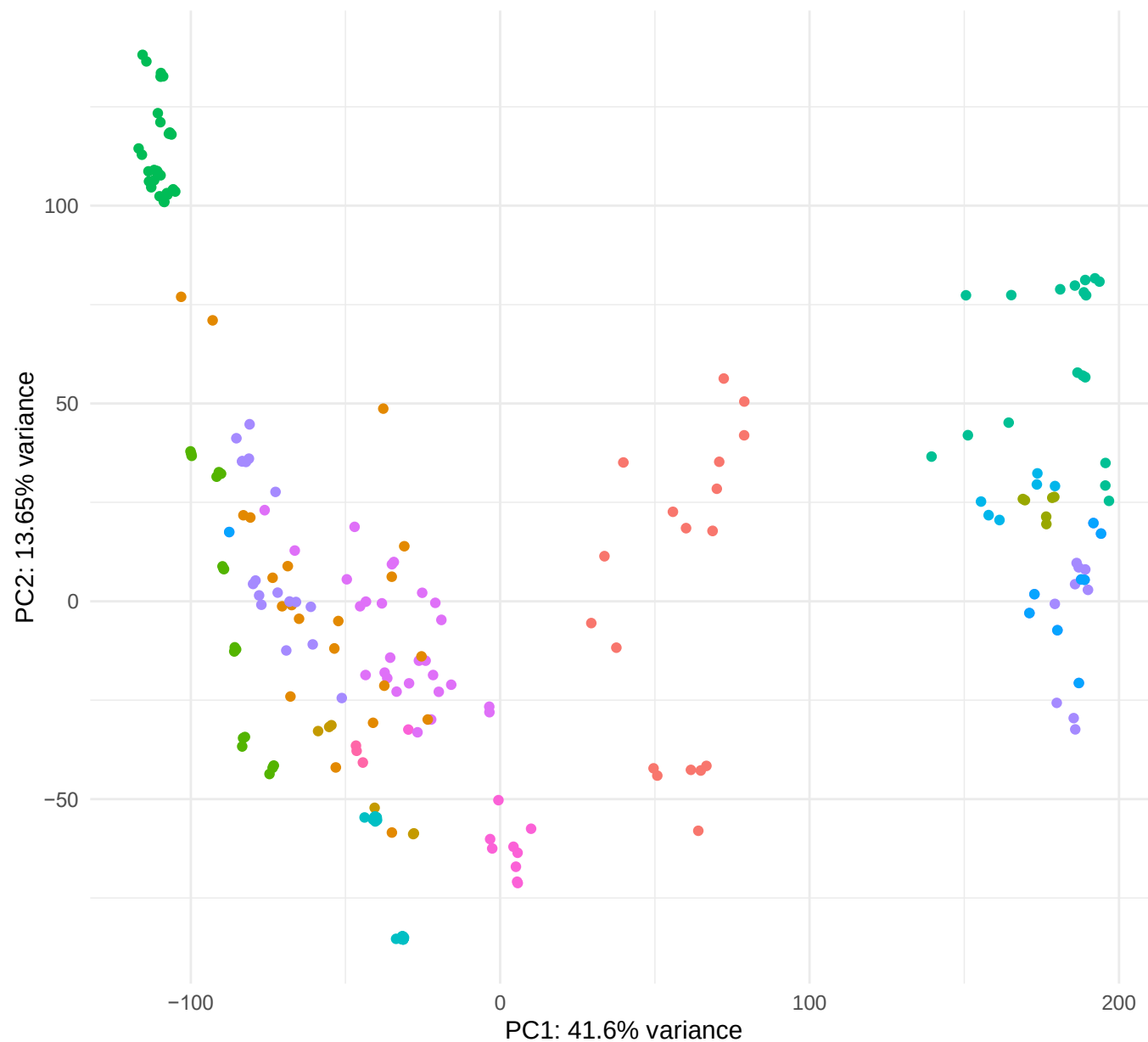
Harder trimming on P.brassicae samples -> change quality threshold to improve quality



Kallisto: Alignment Scores for 283 samples



PCA of filtered vst counts



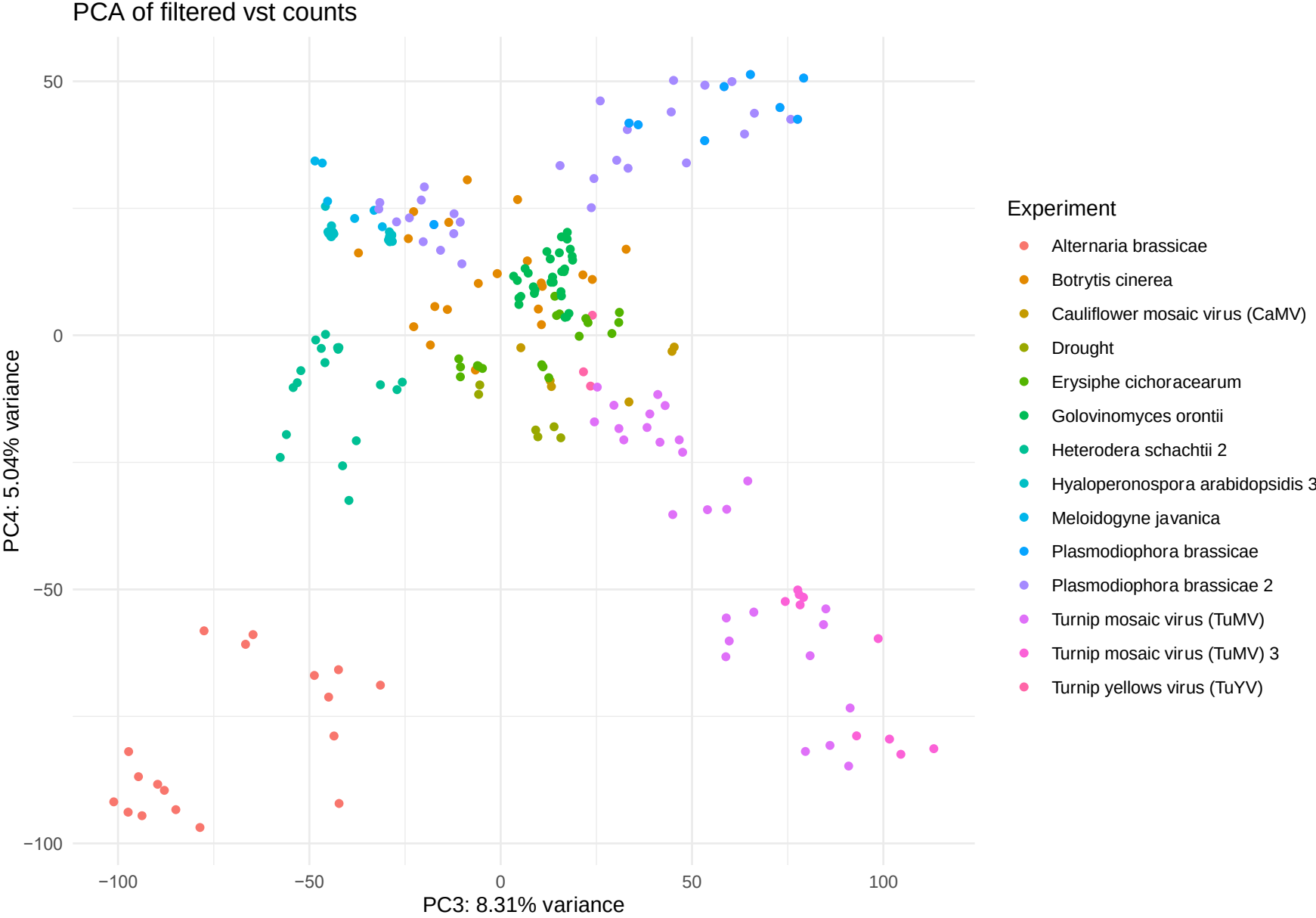
Experiment

- Alternaria brassicae
- Botrytis cinerea
- Cauliflower mosaic virus (CaMV)
- Drought
- Erysiphe cichoracearum
- Golovinomyces orontii
- Heterodera schachtii 2
- Hyaloperonospora arabidopsidis 3
- Meloidogyne javanica
- Plasmodiophora brassicae
- Plasmodiophora brassicae 2
- Turnip mosaic virus (TuMV)
- Turnip mosaic virus (TuMV) 3
- Turnip yellows virus (TuYV)

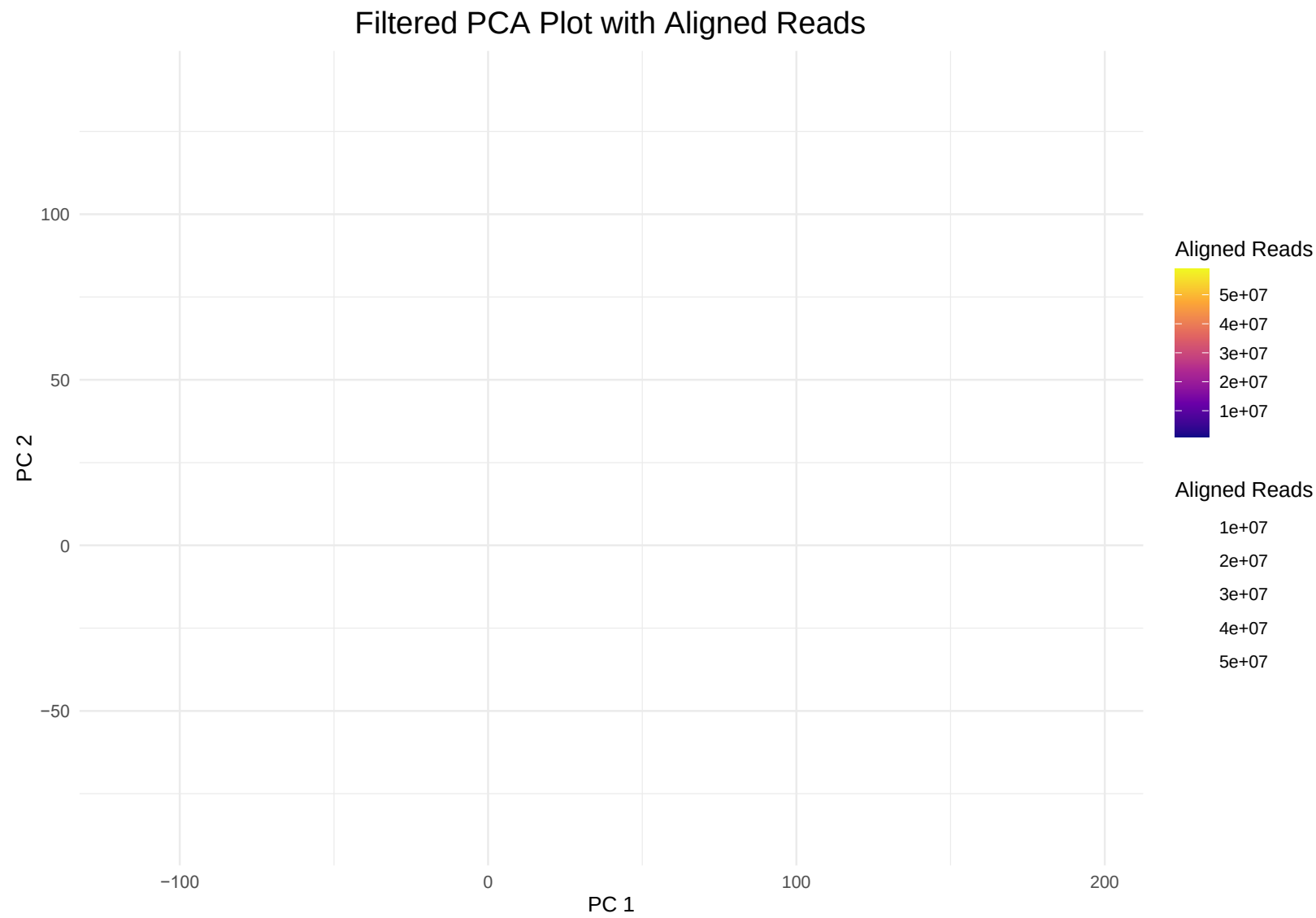
filtering low expressed
genes:
Result:
- before filtering: 27321
genes
- after filtering: 24991
genes

--> keep genes that have
at least a tpm value of
0.5 per sample in at
least 3 of the samples

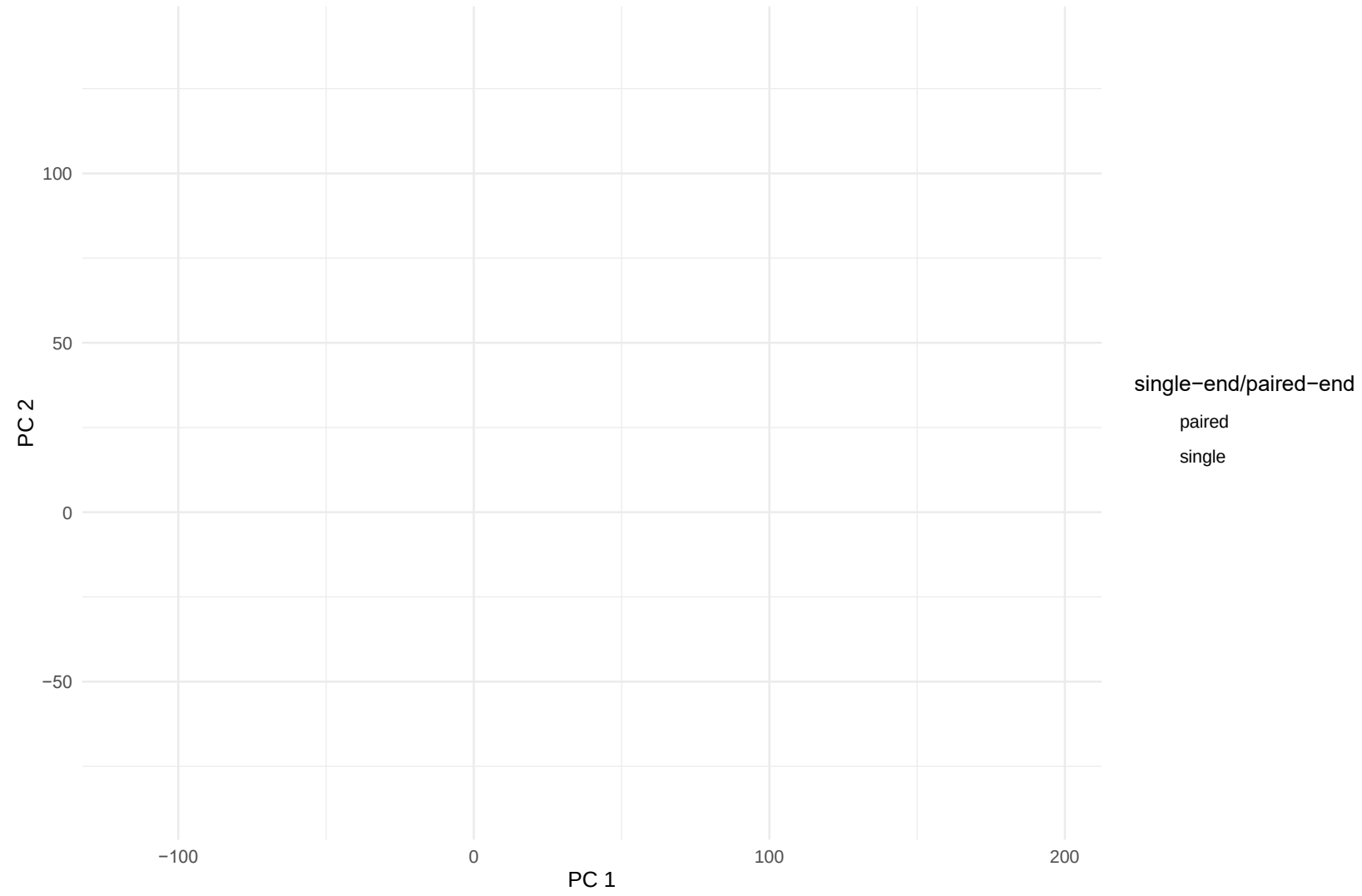
PC 3 and PC 4



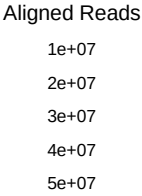
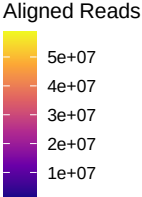
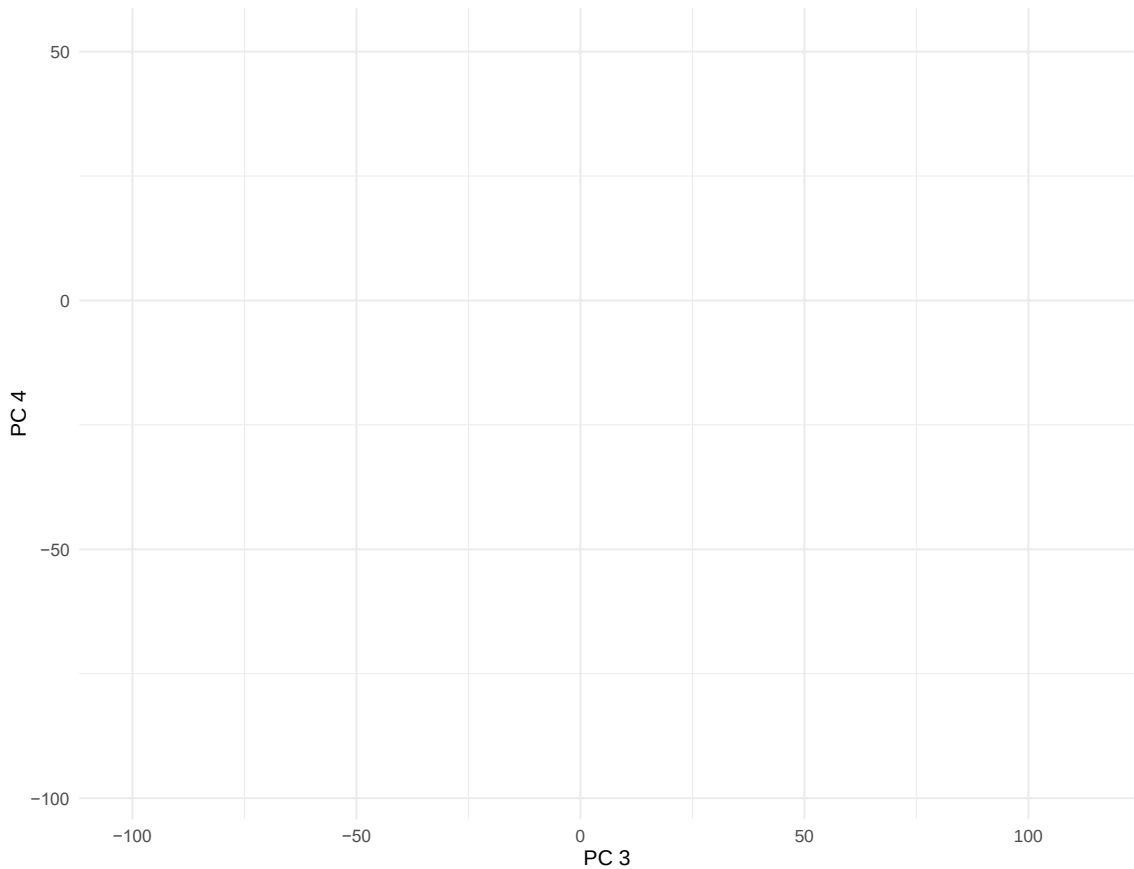
PCA with number of aligned reads per sample



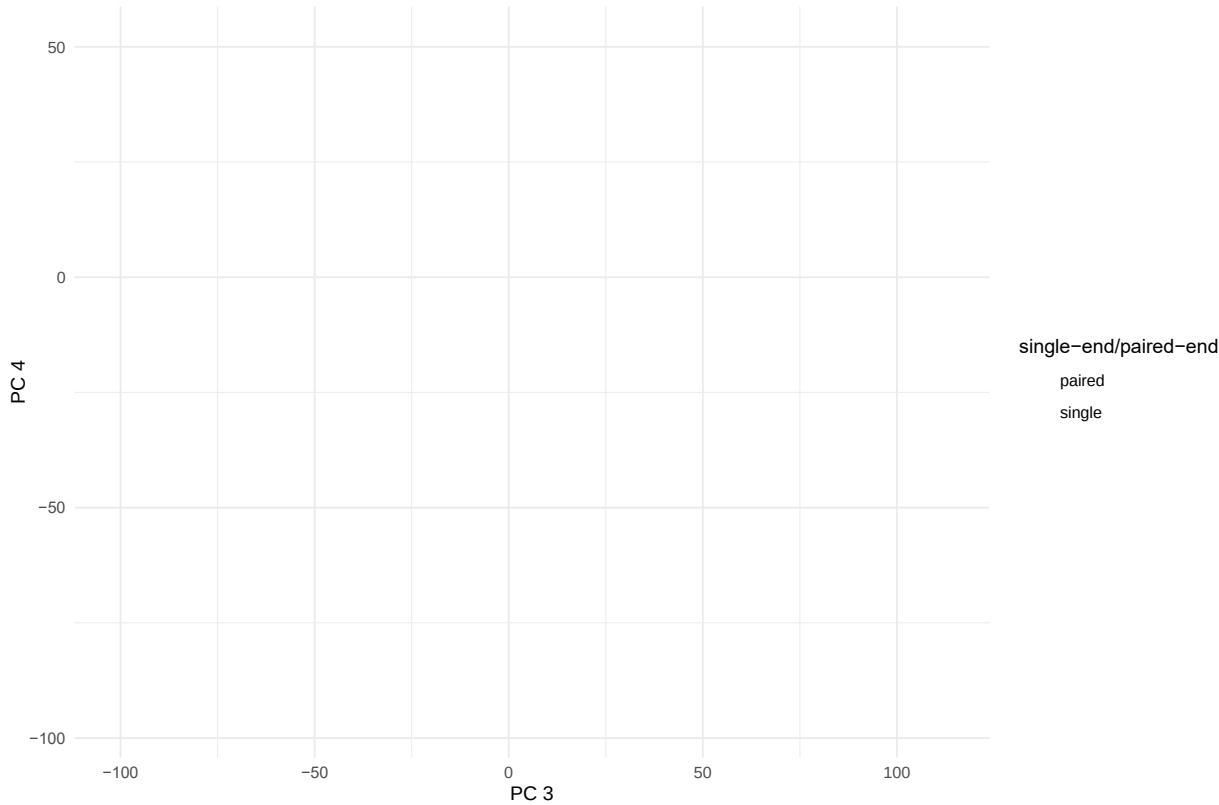
PCA with filtered vst counts



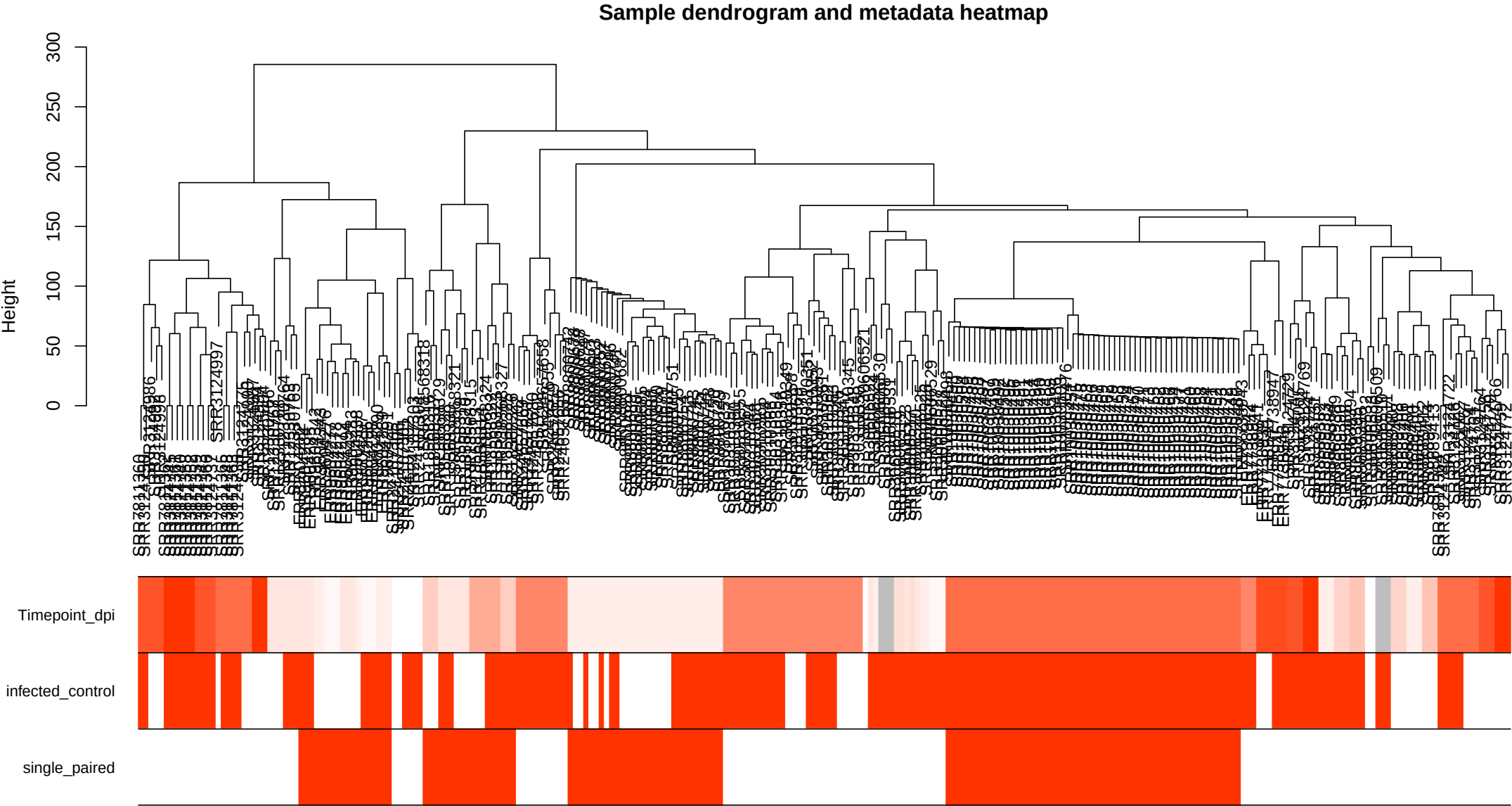
Filtered PCA Plot with Aligned Reads



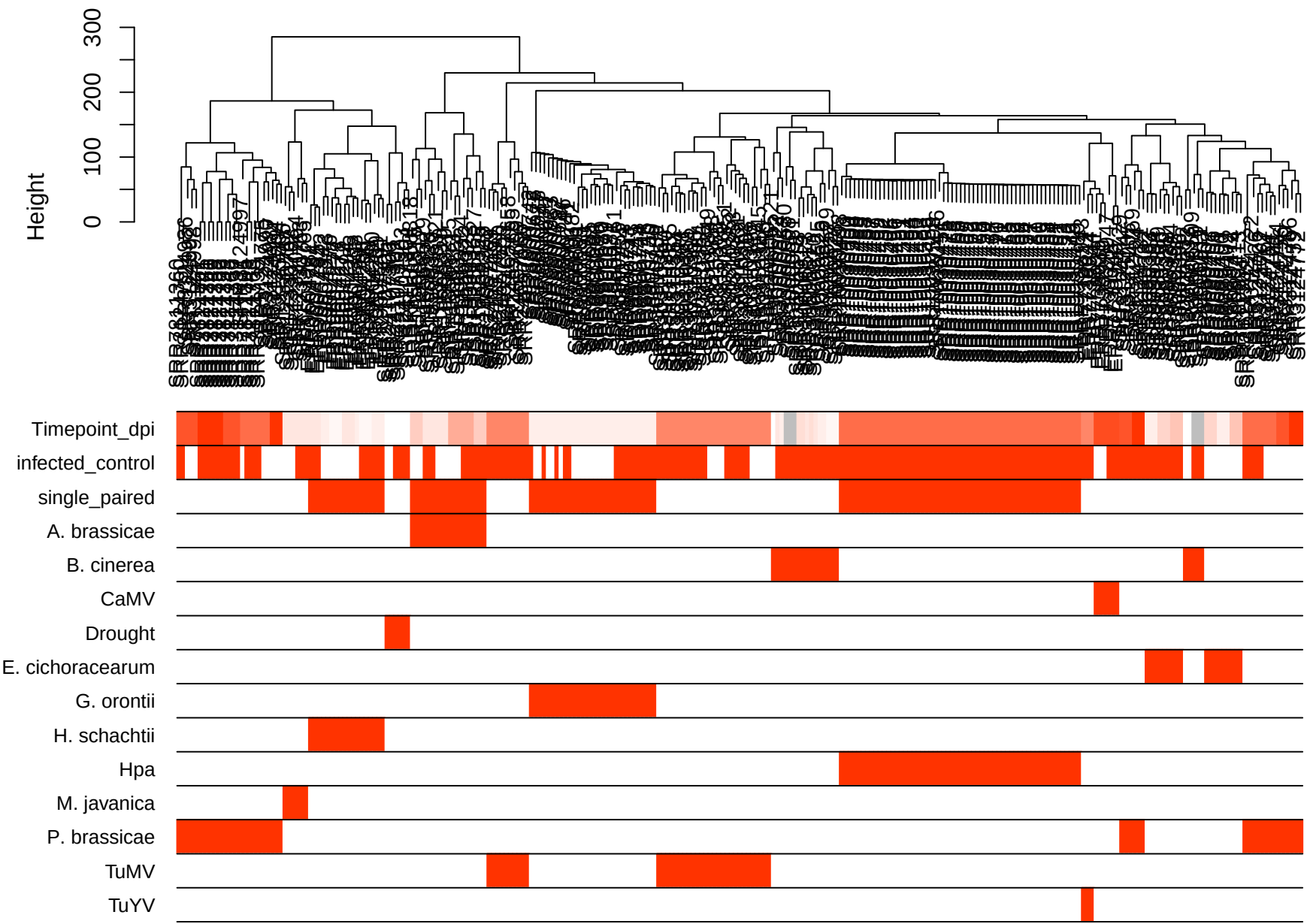
PCA with filtered vst counts



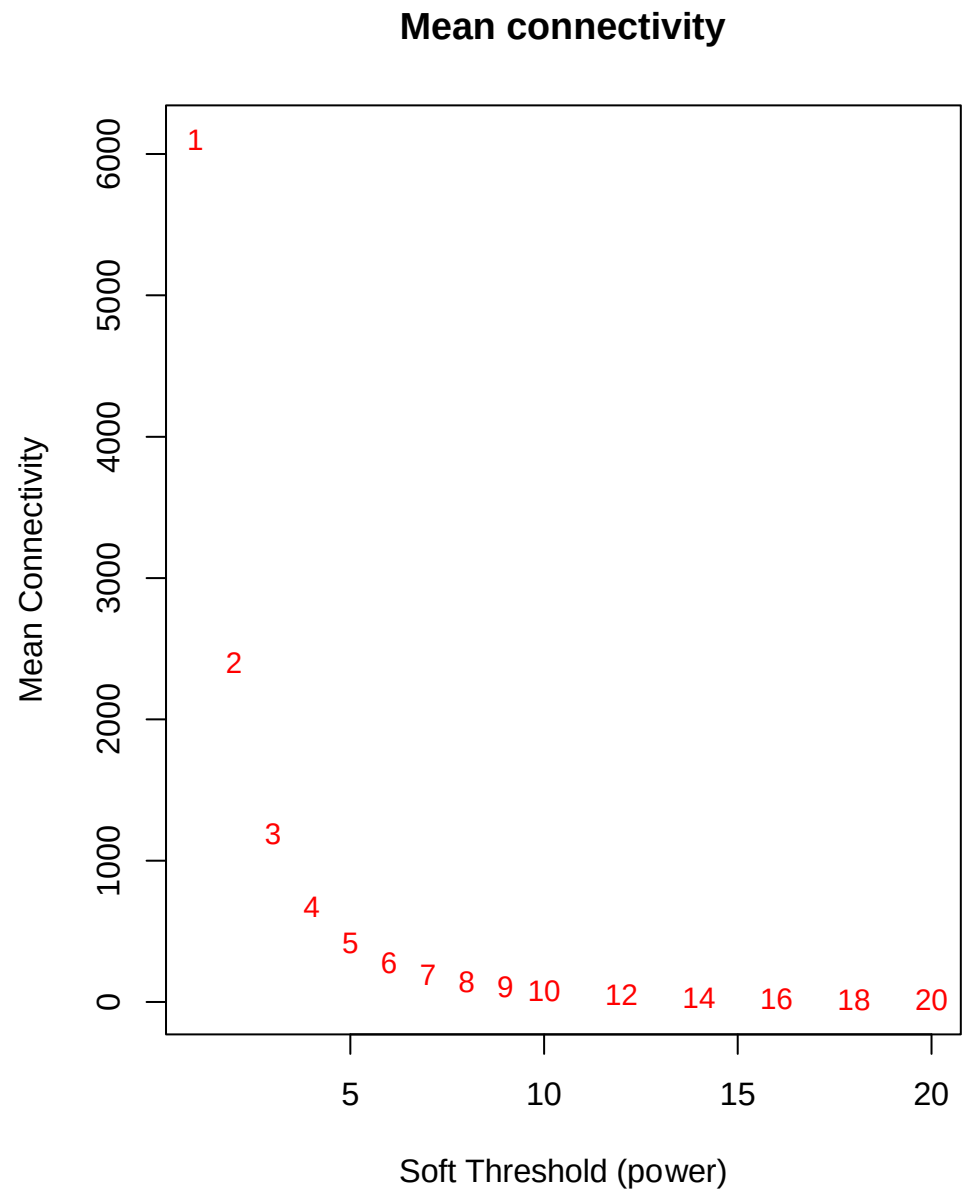
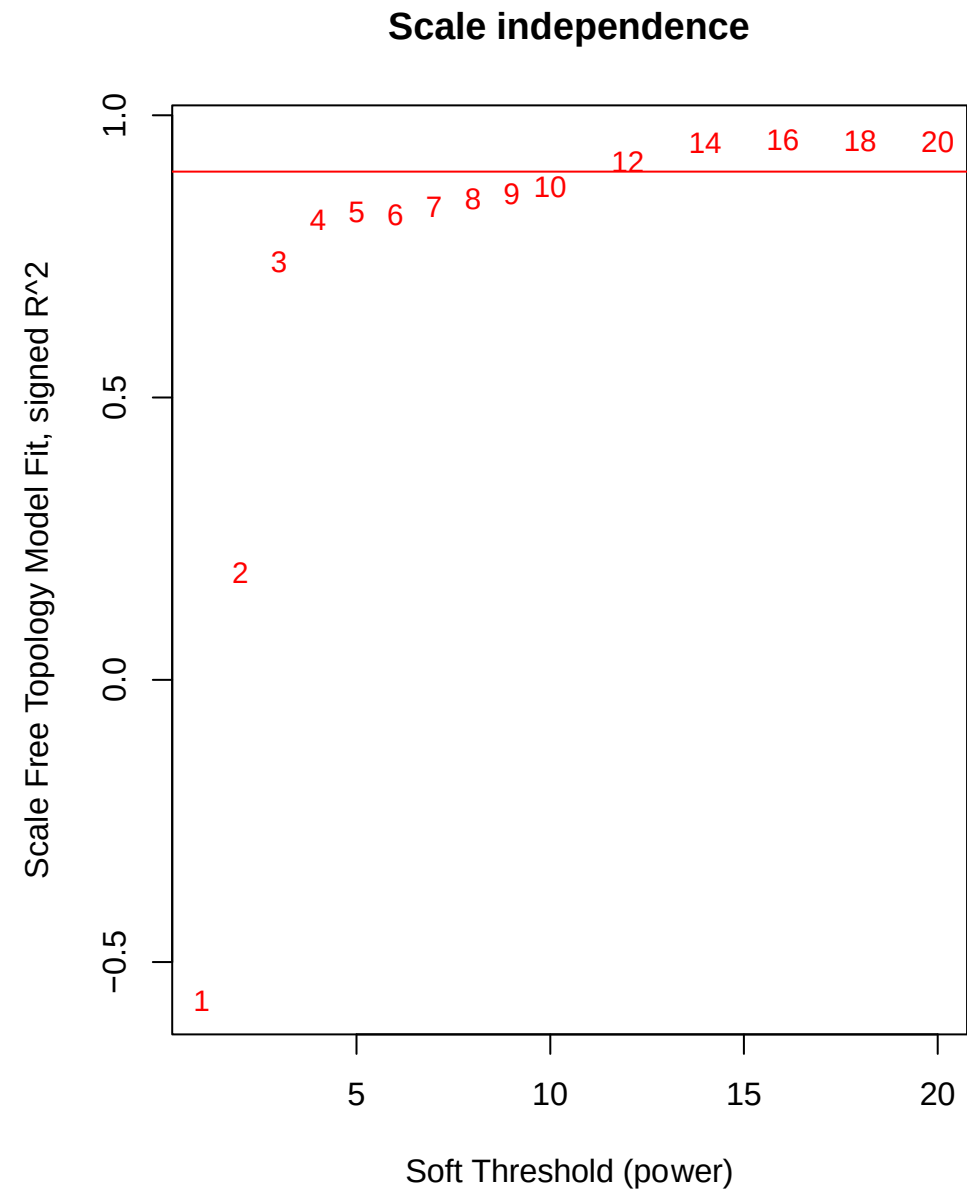
Starting WGCNA:



Sample dendrogram and metadata heatmap



Network construction:

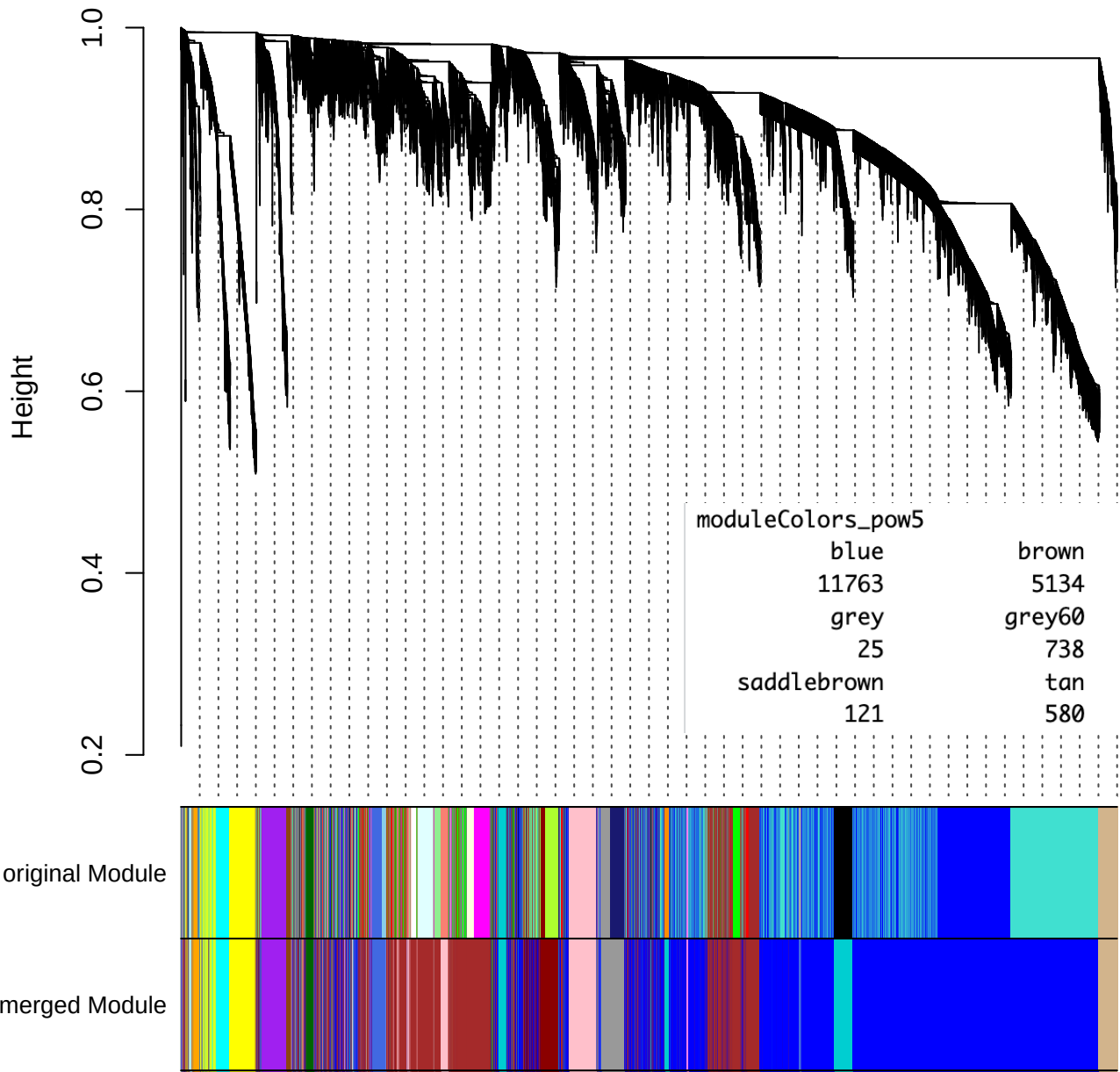


Gene dendrogram and module colors for original and merged module

Soft power: 5

Modules after merging: 18

Genes in grey module: 25

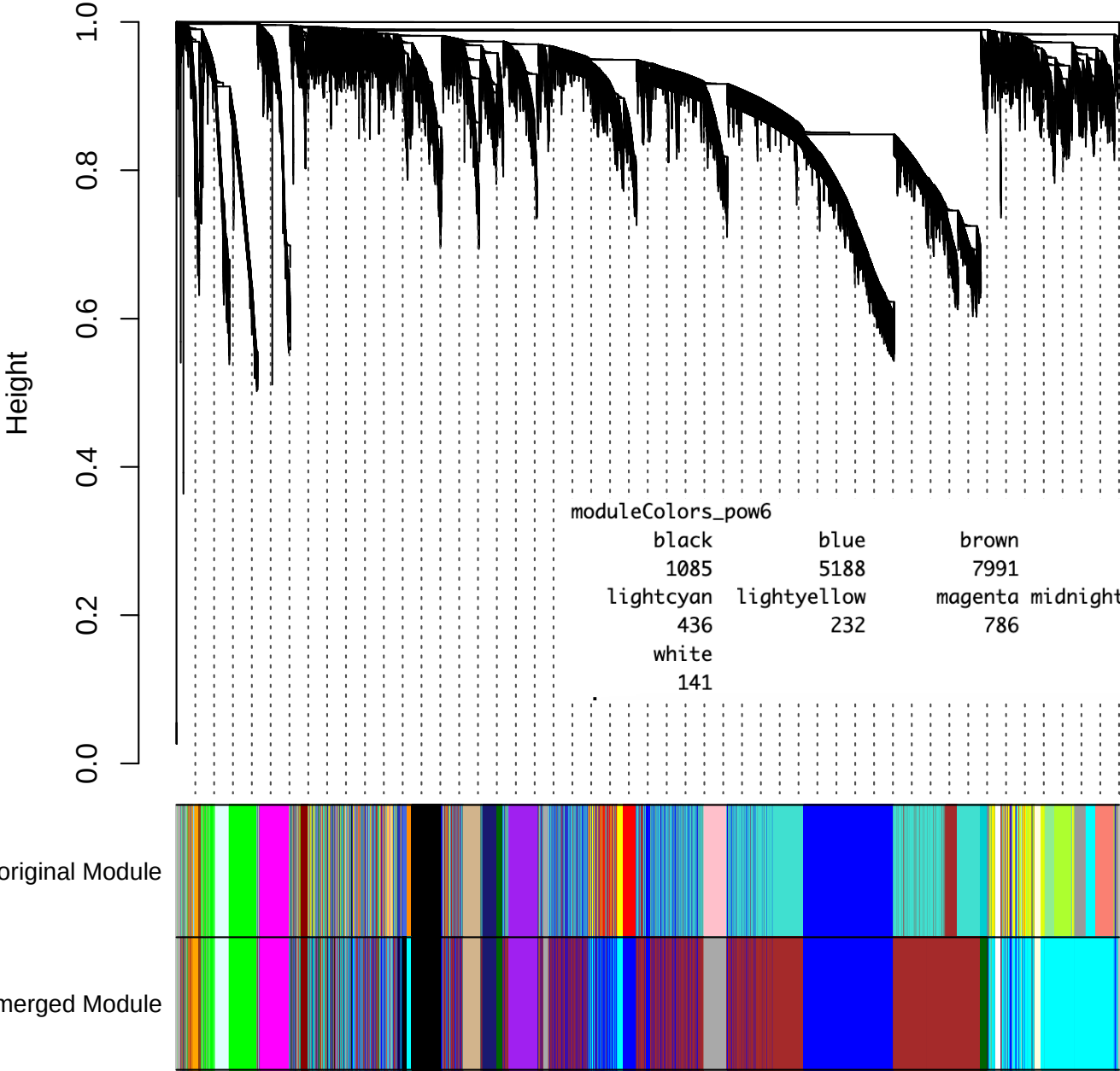


Gene dendrogram and module colors for original and merged module

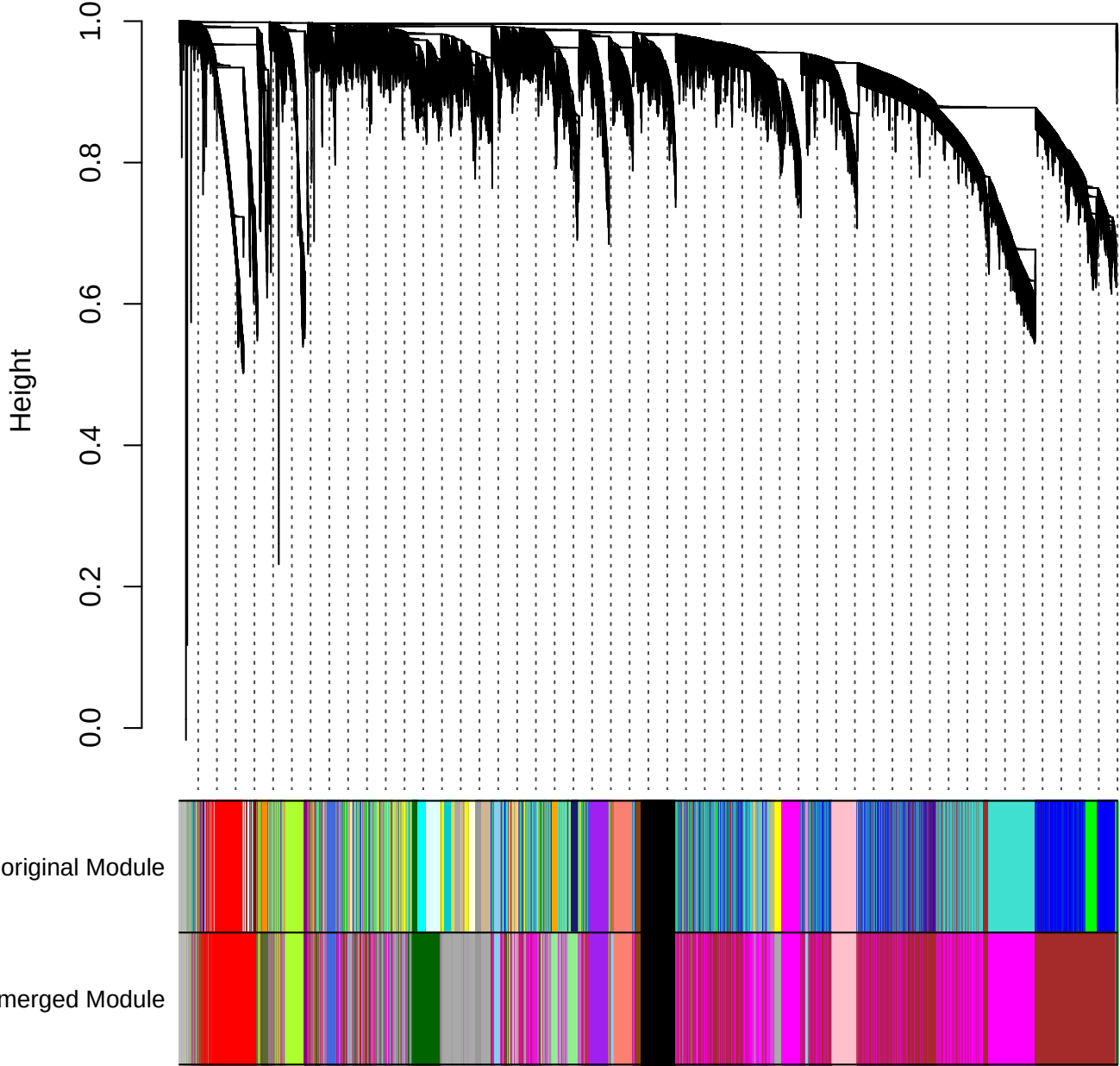
soft power: 6

Modules after merging: 19

Genes in grey module: 131



Gene dendrogram and module colors for original and merged module



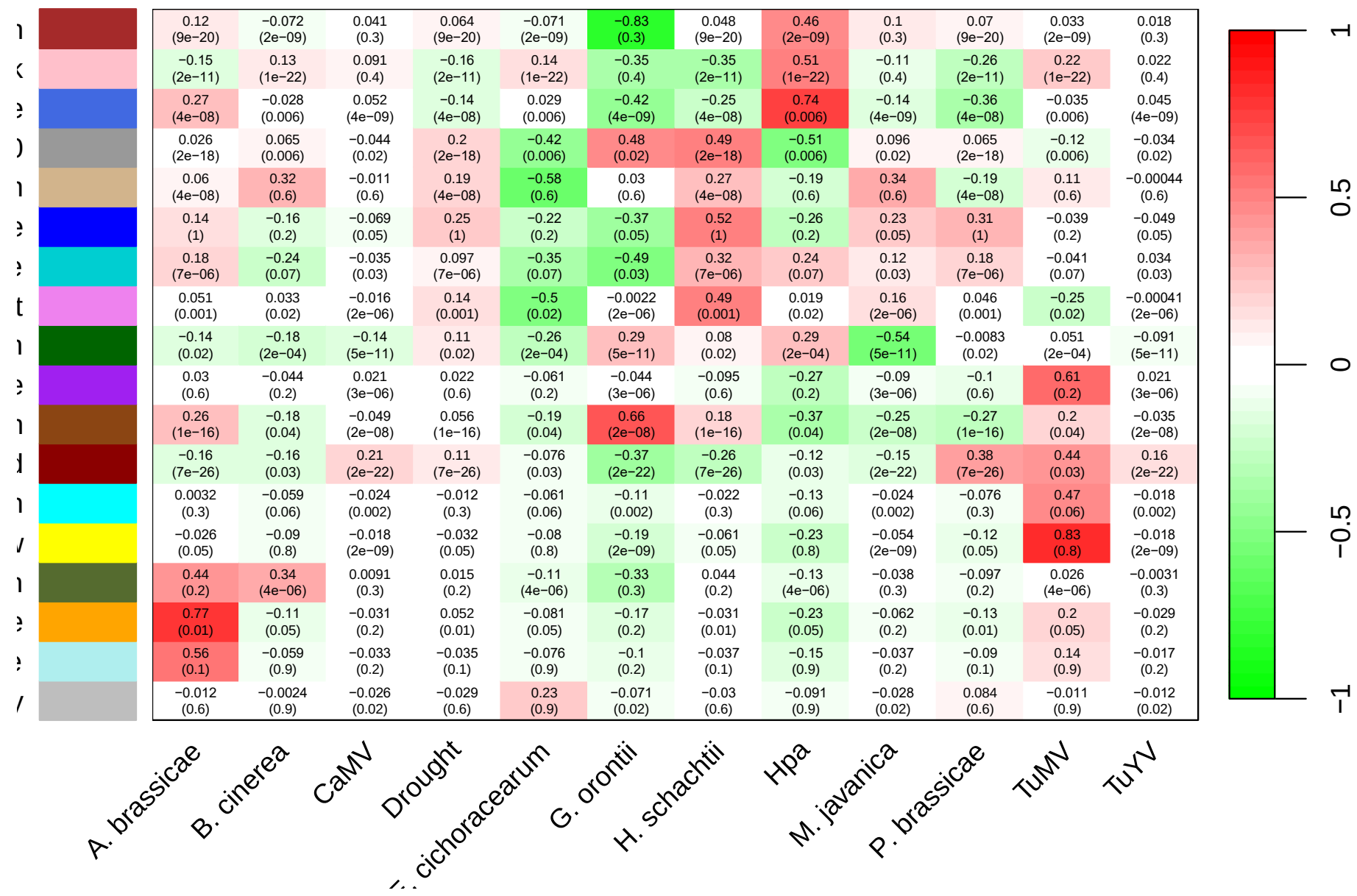
soft power: 7

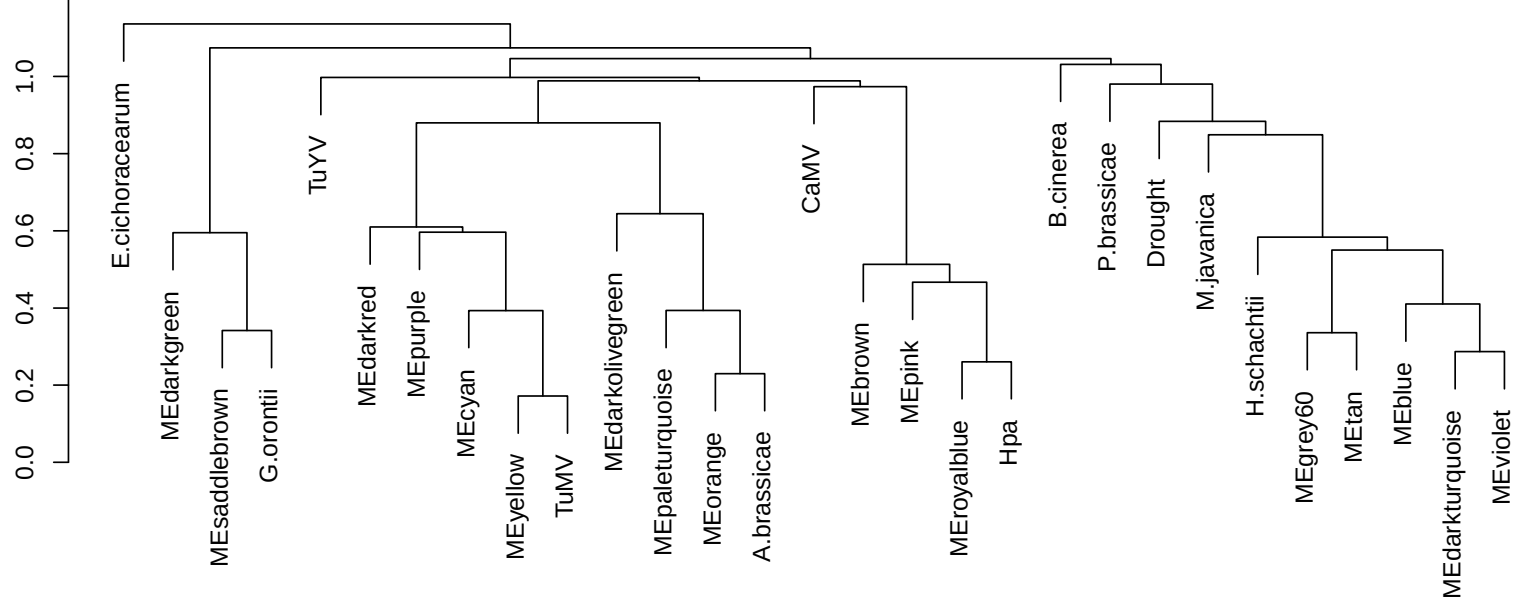
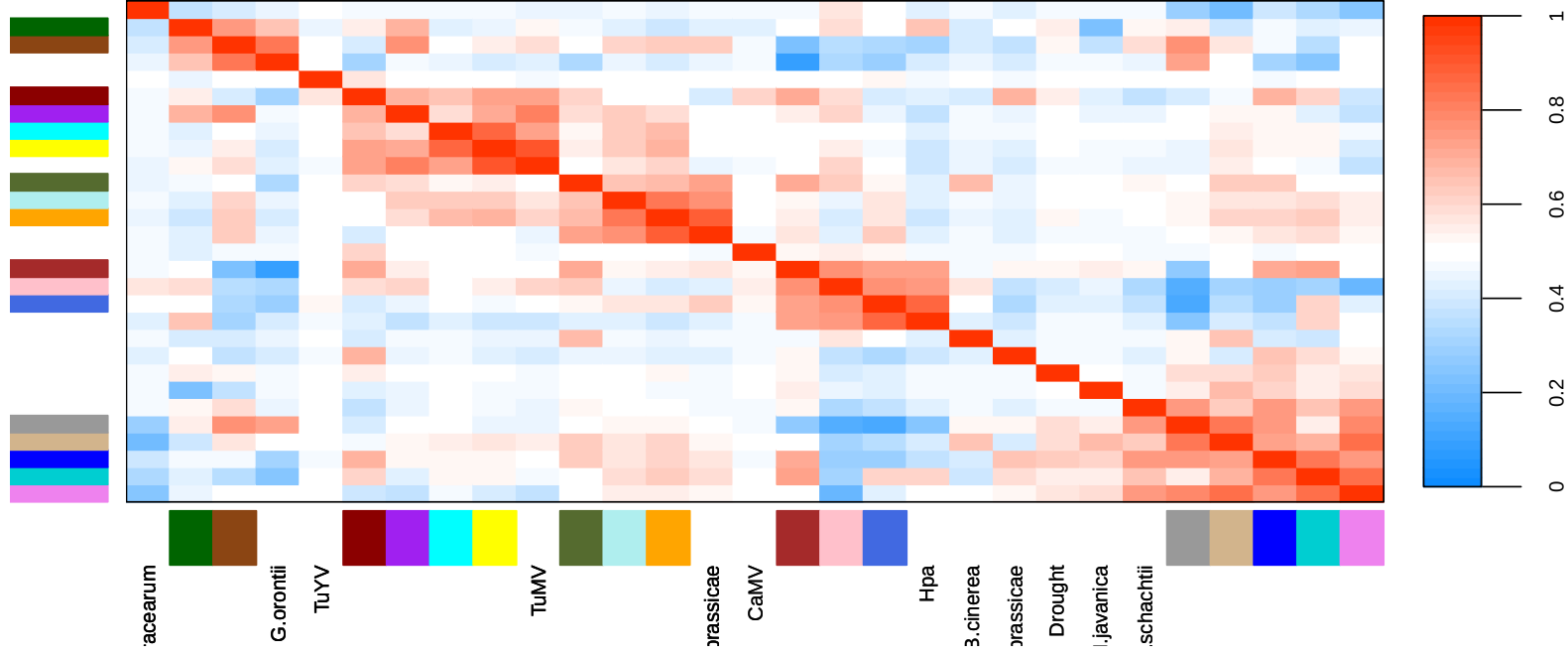
Modules after merging: 19

Genes in grey module: 314

Soft power: 5

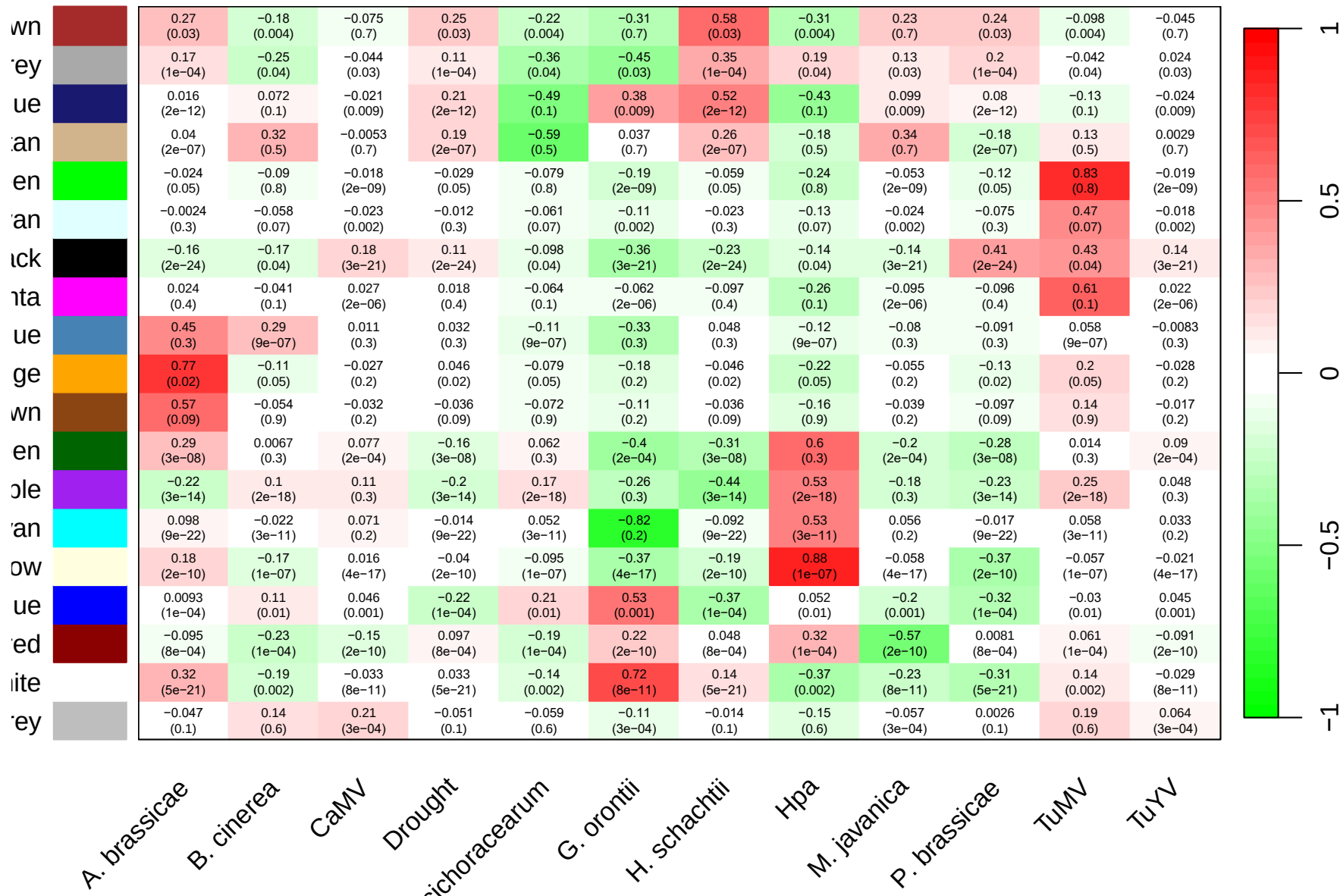
Module–trait relationships with Pathogens

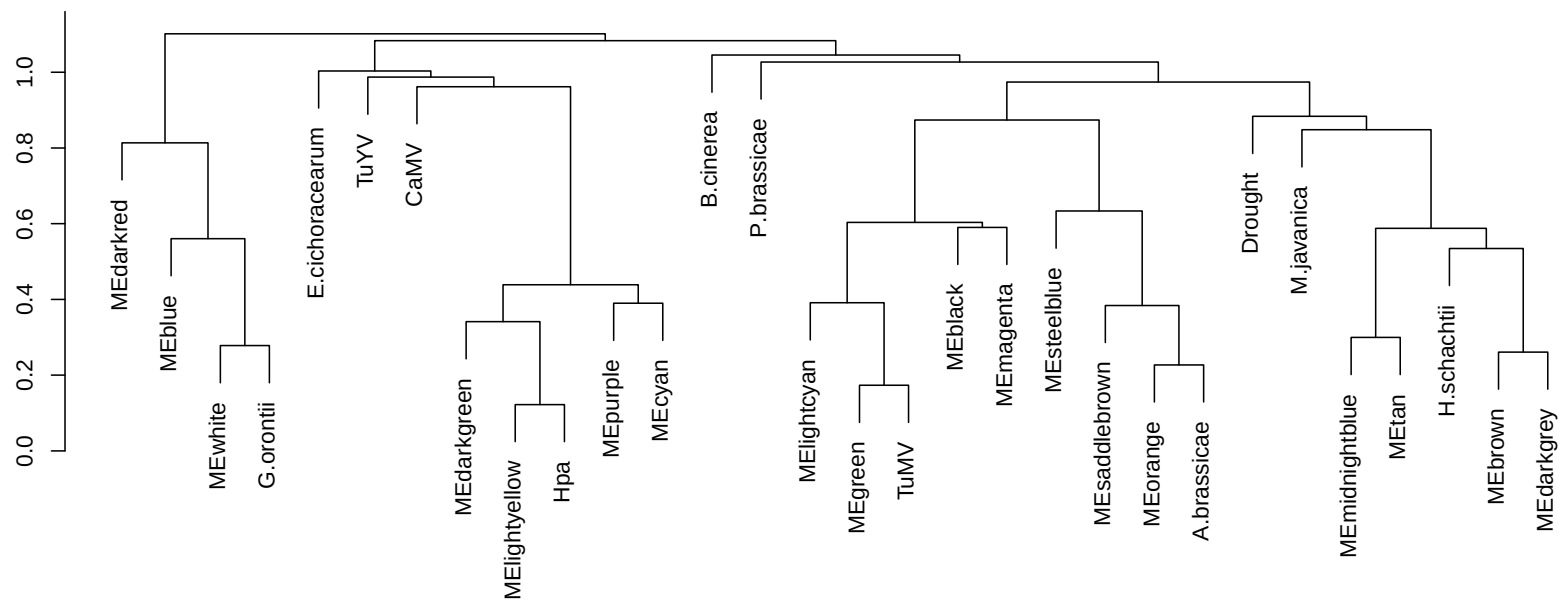




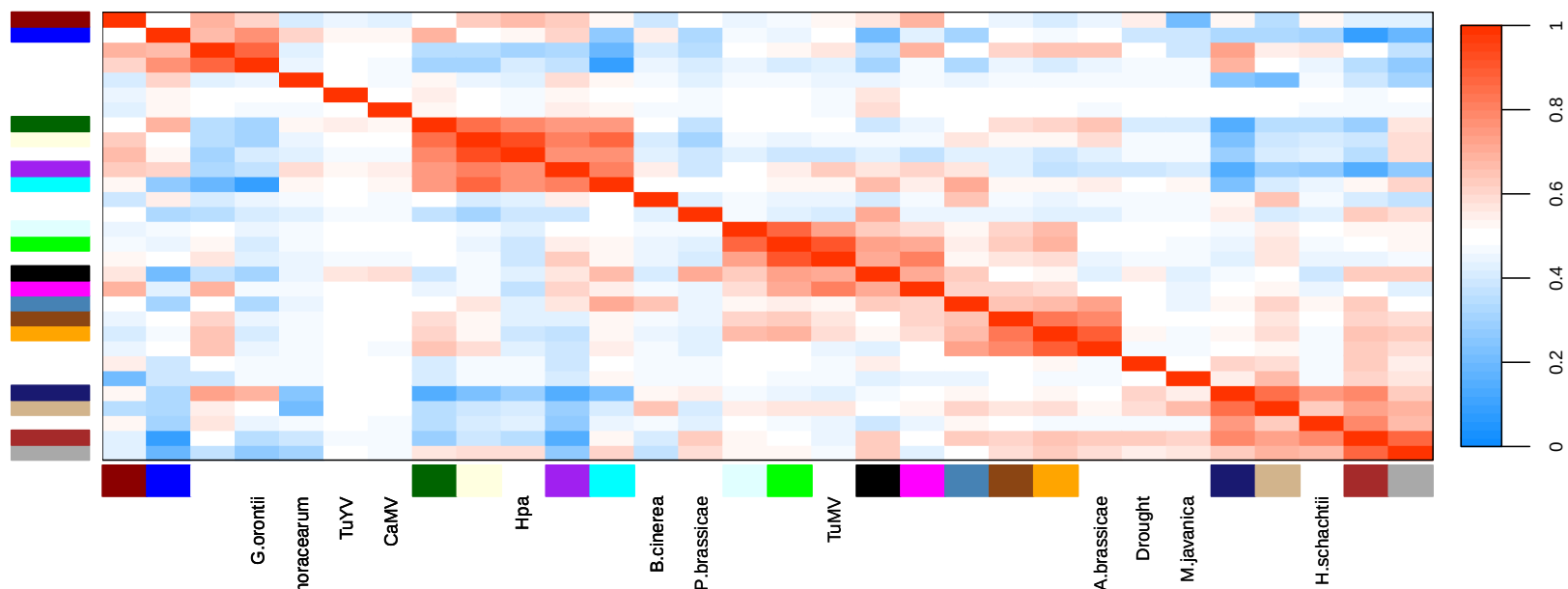
Soft power: 6

Module–trait relationships with Pathogens



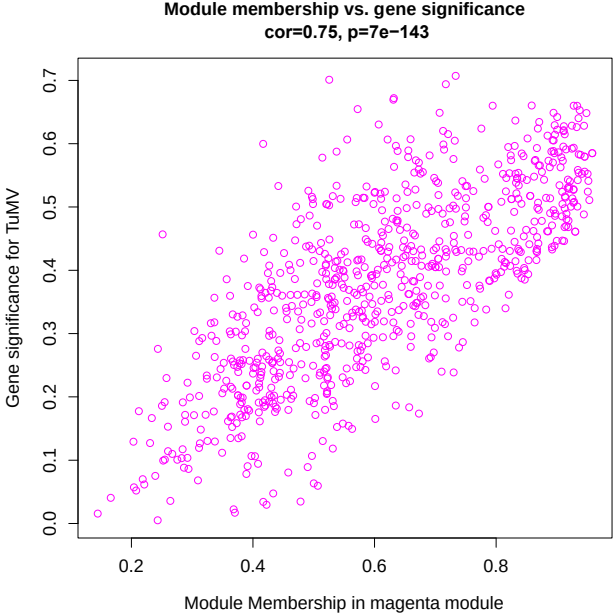
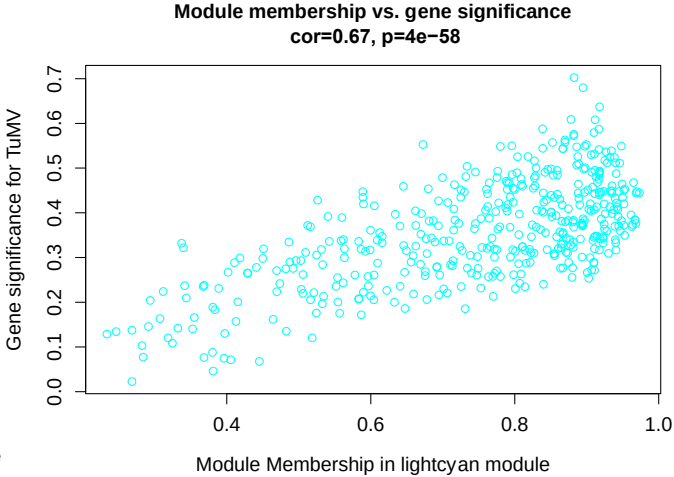
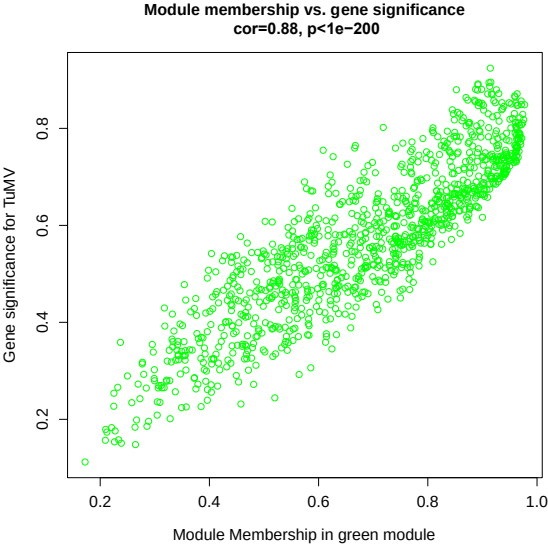
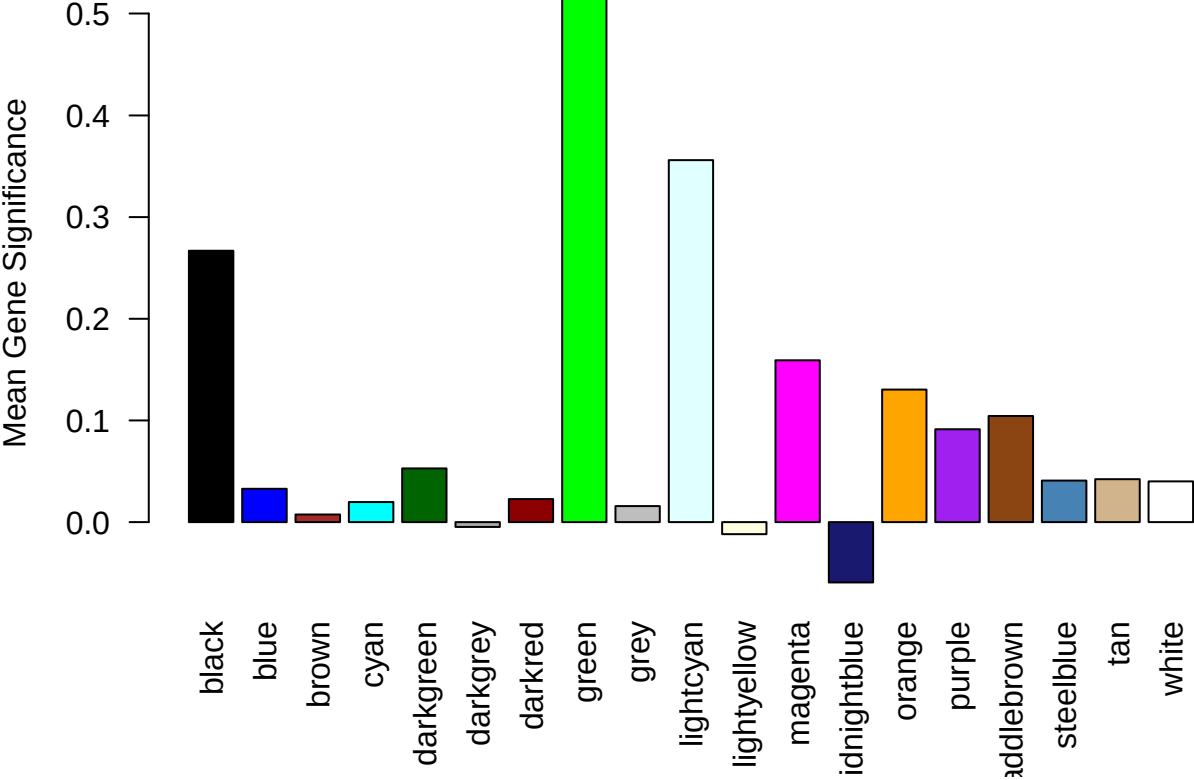


Hierarchical clustering dendrogram of module eigengenes (and all pathogens)
 → Group together eigengenes that are positively correlated



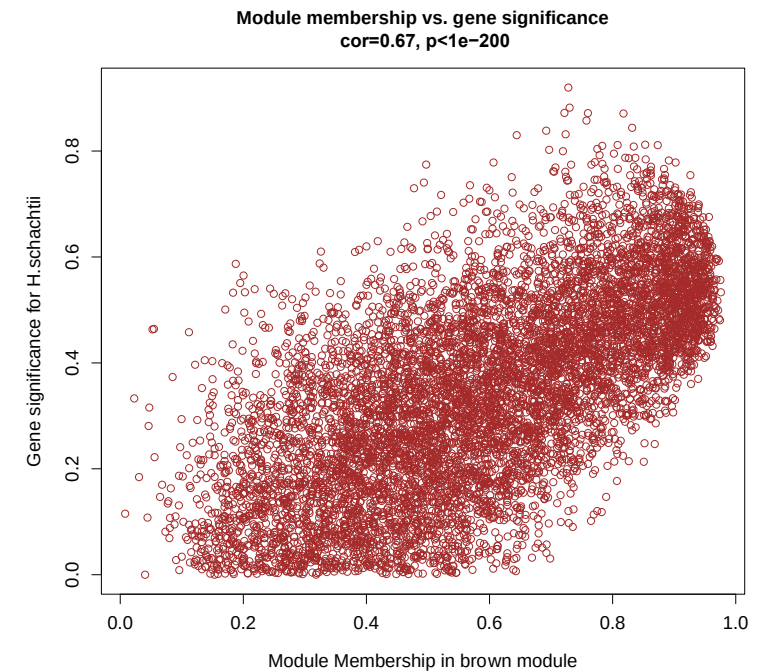
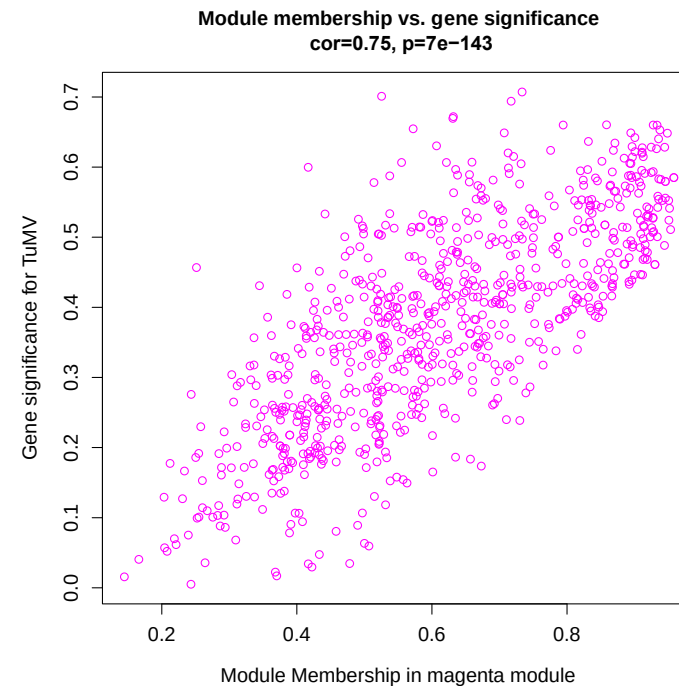
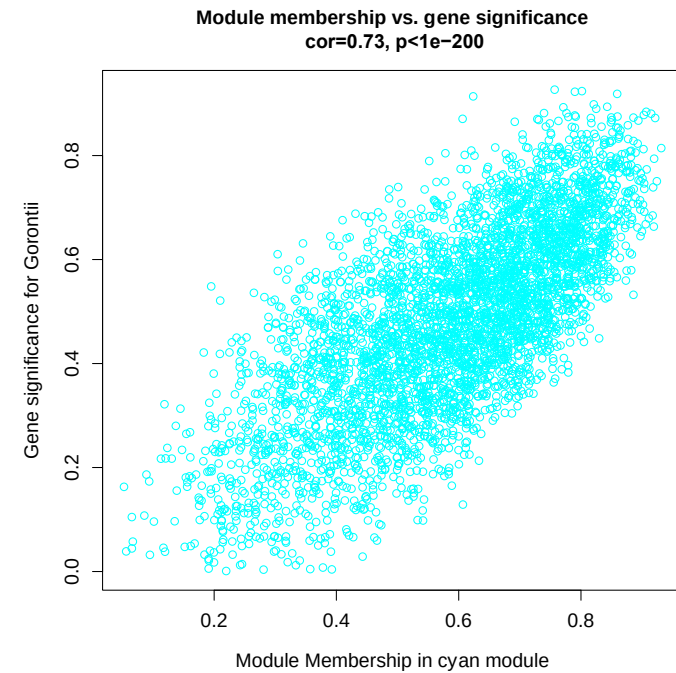
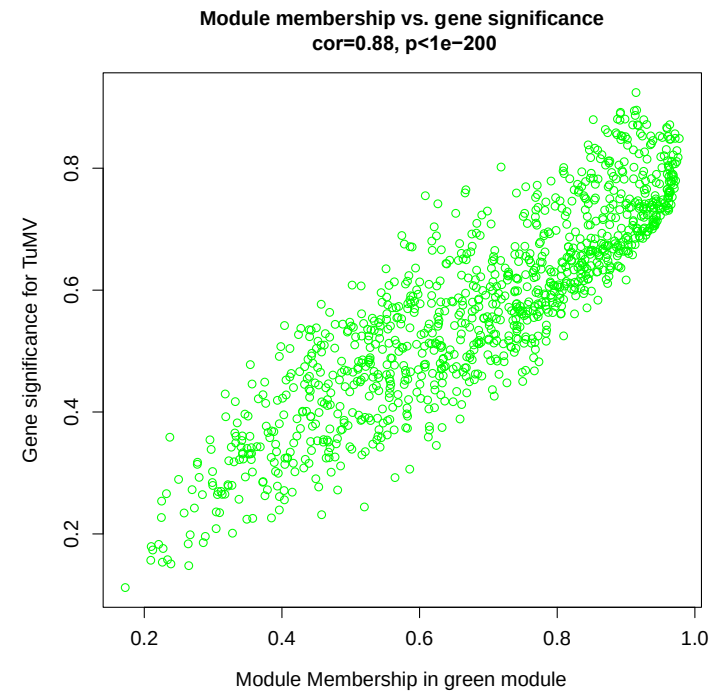
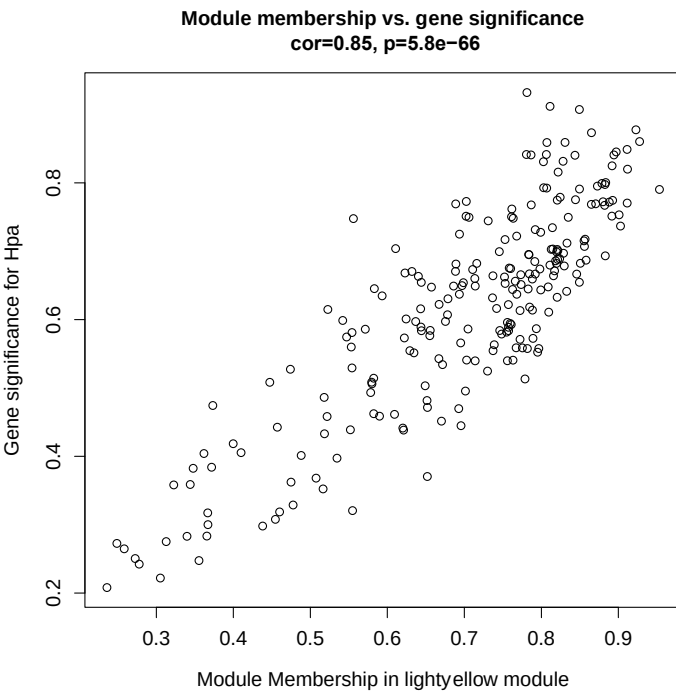
Heatmap of the adjacencies in the eigengene network including all pathogens.

Mean Gene Significance across Modules for TuMV

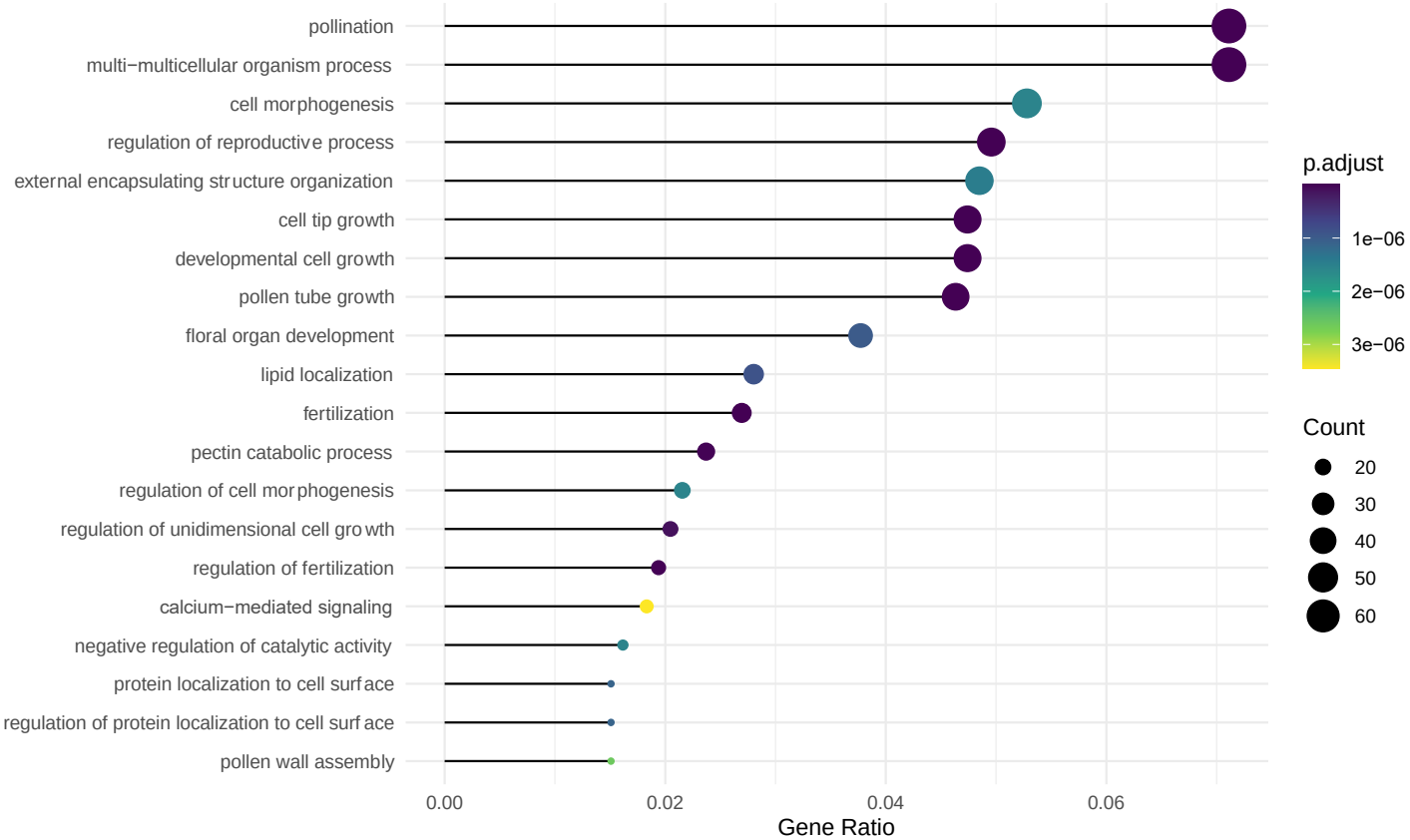


Gene significance and Module Membership

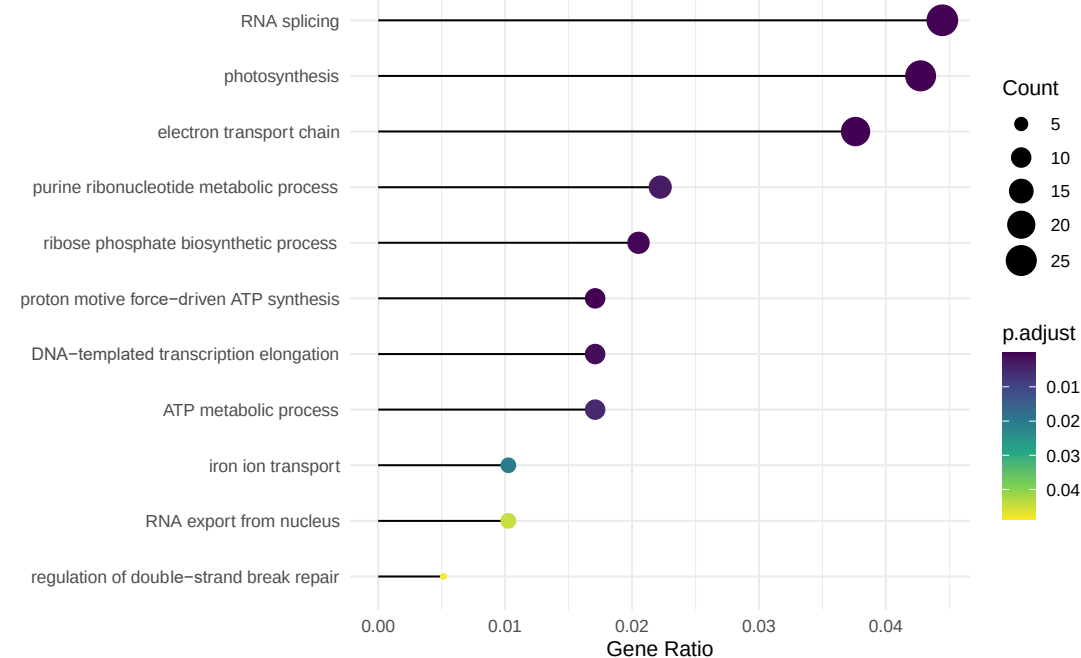
- ➔ Identify genes that have a high significance for the respective pathogen AND
- ➔ High membership in interesting modules



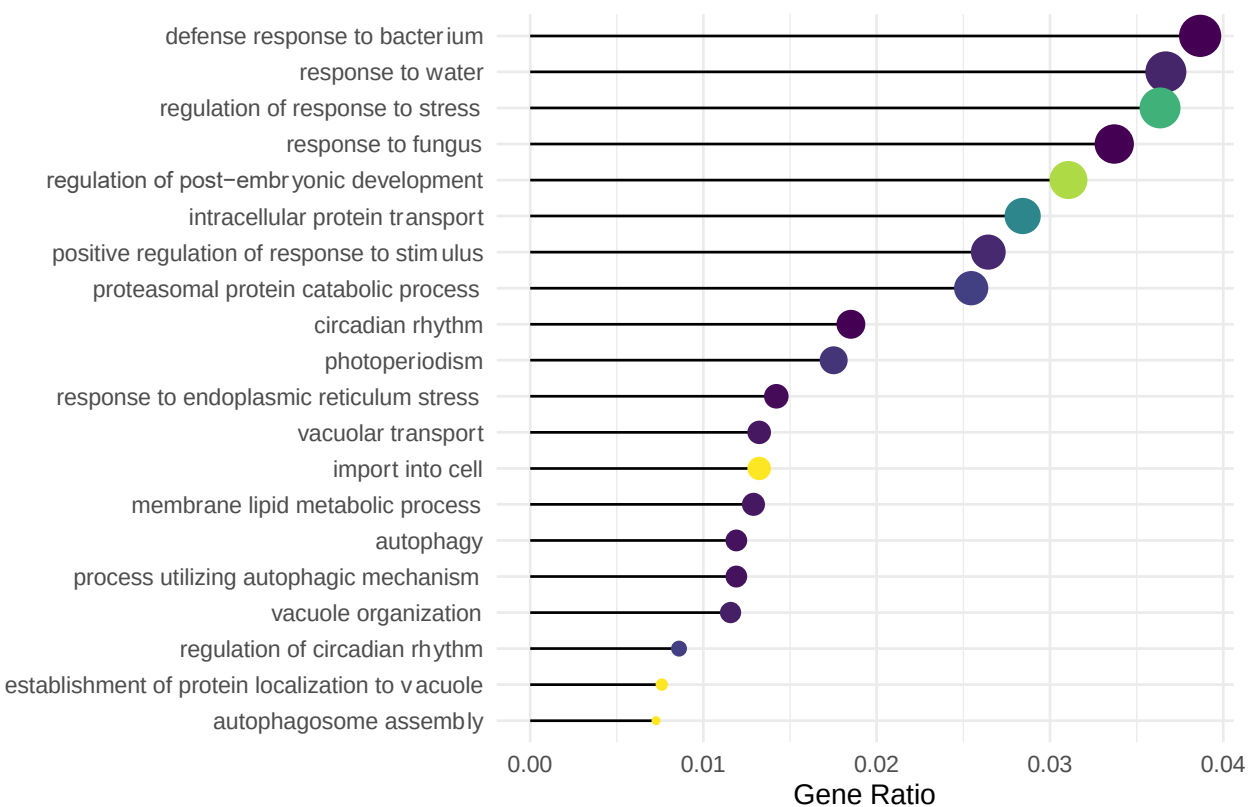
GO Enrichment of green module



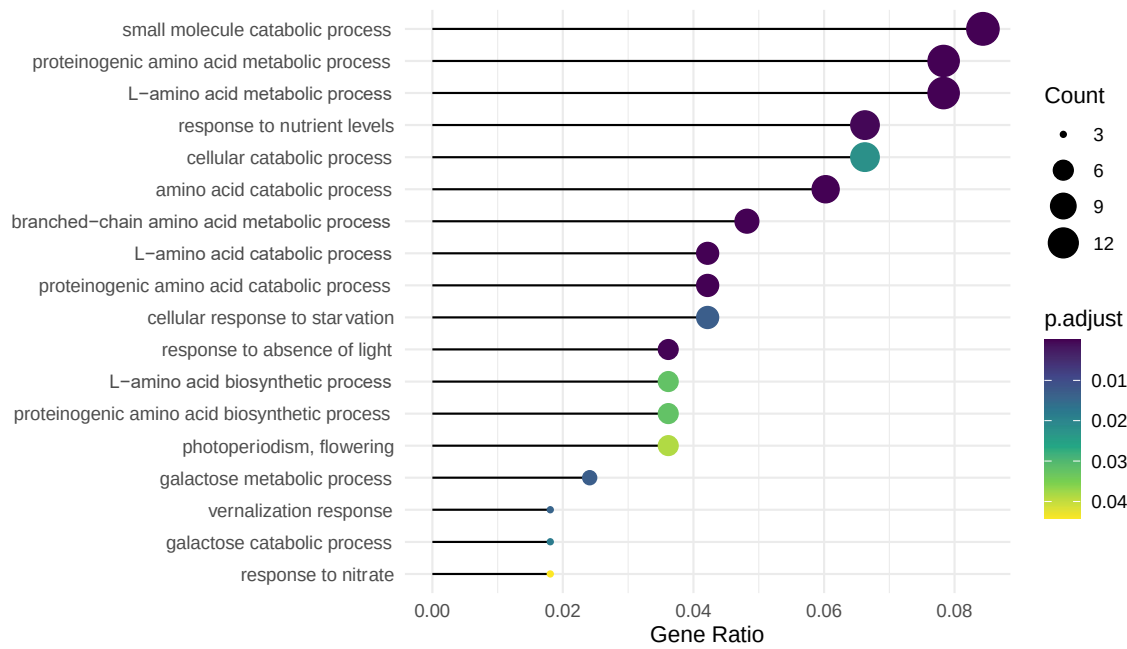
GO Enrichment of magenta module



GO Enrichment of cyan module



GO Enrichment of lightyellow module



GO Enrichment of brown module

